



DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

Galgotias College of Engineering and Technology, Gr. Noida (U.P.)



ARDUINO-BASED HEXAPOD FOR EMERGENCY AND SURVEILLANCE

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Group No. - B9

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TITLE OF PROJECT: ARDUINO-BASED HEXAPOD FOR EMERGENCY AND SURVEILLANCE

SDG GOALS:

SUSTAINABLE DEVELOPMENT GOAL NO.	GOAL	PROJECT ALIGNMENT
SDG 3	Good Health & Well-being	Remote robotic surveillance can reduce the need to expose personnel to potentially hazardous environments, supporting safer emergency/inspection operations.
SDG 9	Industry, Innovation & Infrastructure	Our project integrates robotics, Arduino-based embedded control, custom PCB, sensors, servo actuation and wireless communication into one engineered system.
SDG 11	Sustainable Cities & Communities	The hexapod is designed for emergency surveillance and remote inspection, particularly where human access can be difficult or hazardous.



TITLE: ARDUINO-BASED HEXAPOD FOR EMERGENCY AND SURVEILLANCE

POs:

- PO1:** Application of engineering knowledge in electronics, embedded systems, sensors and mechanical design.
- PO2:** Analysis of obstacle detection, stability, torque, current and power requirements.
- PO3:** Design and development of an autonomous hexapod surveillance robot with integrated sensing.
- PO4:** Testing and investigation of servo, sensor, IMU, power and system performance.
- PO5:** Use of Arduino, ESP32-CAM, PCA9685, HC-SR04, MPU6050 and relevant engineering tools.

PSOs:

- PSO1:** Graduates will be able to design, analyze, and implement electronic circuits, and use techniques relevant to electronics and communication engineering.
- PSO2:** Graduates will demonstrate the ability to apply communication engineering principles in developing innovative and sustainable solutions in areas such as wireless communication systems and antenna design.



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INTRODUCTION

- **Disaster response need:** search-and-rescue teams often have to inspect collapsed, unstable or contaminated areas that are unsafe for people.
- **Why six legs:** a hexapod can keep at least three feet on the ground while walking, so it stays statically stable and can step over rubble and gaps where wheeled robots get stuck.
- **Controller:** the robot is designed around an Arduino Uno R3, which will run the gait algorithm, read the sensors and command every joint.
- **Actuation:** 18 MG996R metal-gear servos (three per leg), driven through two PCA9685 16-channel PWM drivers on a shared I²C bus.
- **Sensing:** an MPU6050 IMU for body tilt and three HC-SR04 ultrasonic sensors (front, left, right) for obstacle detection.
- **Surveillance:** an ESP32-CAM on a pan-tilt mount is planned to stream live video over Wi-Fi.
- **Wiring:** a custom PCB shield replaces roughly 90 loose jumper wires — its design is complete and the Gerber files are ready.



MOTIVATION

- **Uneven environments:** wheeled robots struggle with debris, stairs and uneven ground typical of disaster zones and industrial sites.
- **Operator safety:** sending people into hazardous or physically inaccessible areas carries high risk.
- **Compact embedded platform:** need for a small, self-contained robot that manages its own power and complex locomotion.
- **Remote visual monitoring:** live video lets a team assess a situation safely from a distance.
- **Affordable and reproducible:** commercial inspection and rescue robots are too expensive for student labs; building from off-the-shelf parts keeps the project practical and repeatable.



PROBLEM STATEMENT

1. Multi-legged locomotion with 3 degrees of freedom (DOF) per leg for better terrain adaptability.
2. Embedded control of 18 heavily loaded servo motors at the same time without signal degradation.
3. Stable, multi-rail power distribution that isolates logic components from high-current actuator spikes.
4. Remote visual surveillance through a pan-tilt camera over Wi-Fi.
5. Reliable wiring on a machine that vibrates with every step.



OBJECTIVES

1. To design and develop a low-cost six-legged (hexapod) robot using an Arduino Uno and servo motors that can walk stably over flat and moderately uneven surfaces.
2. To integrate ultrasonic distance sensing and an ESP32-CAM module so that the robot can perform basic obstacle avoidance and stream live video for surveillance.
3. To design and evaluate a control scheme that coordinates the leg movements and sensor feedback for safe navigation in emergency scenarios.



LITERATURE REVIEW

S. No.	REFERENCE	FINDINGS	LIMITATION / RELEVANCE TO OUR PROJECT
1	Ji et al. — IEEE Access, 2024 (Base paper)	RL-based gait and foot-trajectory planning; static stability margin used as the stability measure; tilt-based gait switching.	Needs RL training (PyBullet) and a GPU — not feasible on an Arduino. We adopt only the stability concepts.
2	Mohamed et al. — Proc. IMechE Part C, 2024	Kinematic model and body levelling of a hexapod with telescopic legs.	Telescopic actuators exceed our budget; the modelling approach guided our leg torque check.
3	Liu et al. — J. Field Robotics, 2025	Crab-like lateral walking with hierarchical control on unknown rugged terrain.	Long lateral walking loads the hip joints heavily; we start with a forward tripod gait.
4	Xu et al. — J. Field Robotics, 2024	Improved knee joint raises obstacle clearance without extra actuators.	Shows the effect of leg geometry; our knee uses a standard rotary servo.



LITERATURE REVIEW (CONTD.)

S. No.	REFERENCE	FINDINGS	LIMITATION / RELEVANCE TO OUR PROJECT
5	Saengsint et al. — IEEE ROBIO, 2023	Rescue hexapod with AI-based human detection and tracking.	On-board AI needs far more processing than a microcontroller.
6	Kulkarni et al. — Springer LNEE, 2024	Arduino Uno + ultrasonic sensor + ESP32-CAM surveillance robot with live video.	Wheeled platform; confirms the sensing and camera stack we are using.
7	Mahmud et al. — IEEE Access, 2024	Synthetic images used to improve victim detection in disaster scenes.	Shows the value of a visual link; victim detection is outside our scope.



LITERATURE REVIEW (CONTD.)

S. No.	REFERENCE	FINDINGS	LIMITATION / RELEVANCE TO OUR PROJECT
8	Cruz Ulloa et al. — Applied Sciences, 2024	Compares thermal, multispectral and RGB vision for victim detection.	Thermal / multispectral cameras exceed our budget; RGB (ESP32-CAM) chosen.
9	Jadeja et al. — Scientific Reports, 2024	Deep-learning survivor detection on a snake robot.	Needs neural-network hardware; detection is better done on the operator side.
10	Gelfert — RiTA 2022 (Springer), 2023	Real-time victim detection in smoke using a multi-sensor unit.	Specialised sensors and deep learning; beyond this project.



BASE PAPER FINDINGS

Base paper: S. Ji, W. Wei, X. Liu and J. Wu, “Optimization Control of Attitude Stability for Hexapod Robots Based on Reinforcement Learning,” IEEE Access, vol. 12, 2024.

- **Static stability:** the robot is statically stable when the projection of its centre of mass (CoM) lies inside the support polygon formed by the feet on the ground.
- **Stability margin (SSM):** the shortest distance from the CoM projection to the polygon edges — a larger margin means safer walking on rough ground.
- **Hierarchical RL control (RL-FI-RW):** a policy network outputs only four gait parameters (X, Y, lift height H, rotation Ω); inverse kinematics and a trajectory controller drive the joints.
- **Tilt-based gait switching:** tripod gait normally, 4-legged gait above 20° body inclination and 5-legged gait at 25° .
- **Results (simulation only):** in PyBullet, with a 15 kg robot model, the method showed less body sway and longer travel than CPG-based methods on rough terrain.



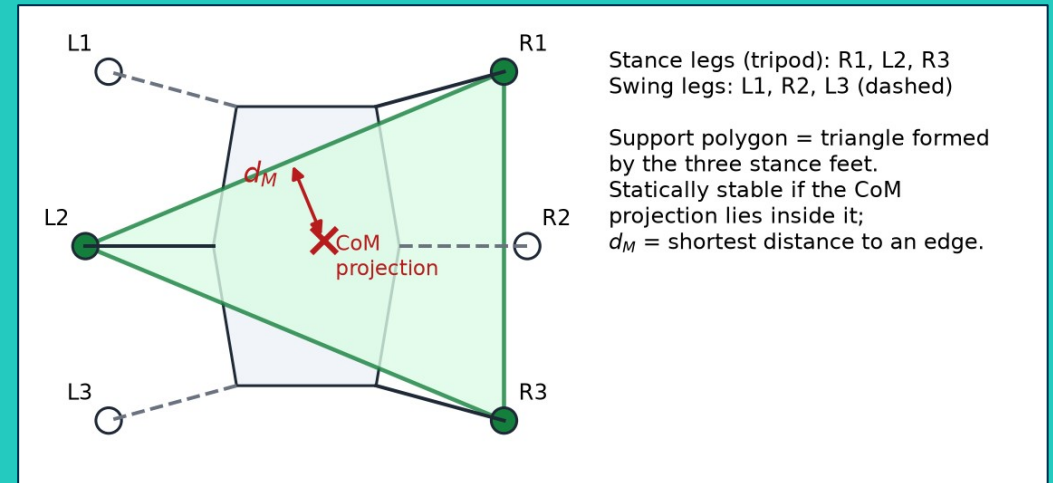
BASE PAPER — RELEVANCE TO OUR PROJECT

What we use from the paper

- Support polygon and stability margin → tripod gait, and a low, central centre of mass (battery and converters under the body plate).
- Body-inclination feedback → MPU6050 IMU in our design.
- More legs on the ground for rough terrain → planned 4-legs-down gait.
- Smooth swing trajectories instead of abrupt joint moves.

What we do not use

- The reinforcement-learning controller — it needs GPU training and cannot run on an Arduino Uno (2 KB RAM).
- Research gap addressed: applying the stability concepts to a low-cost, microcontroller-based physical robot.



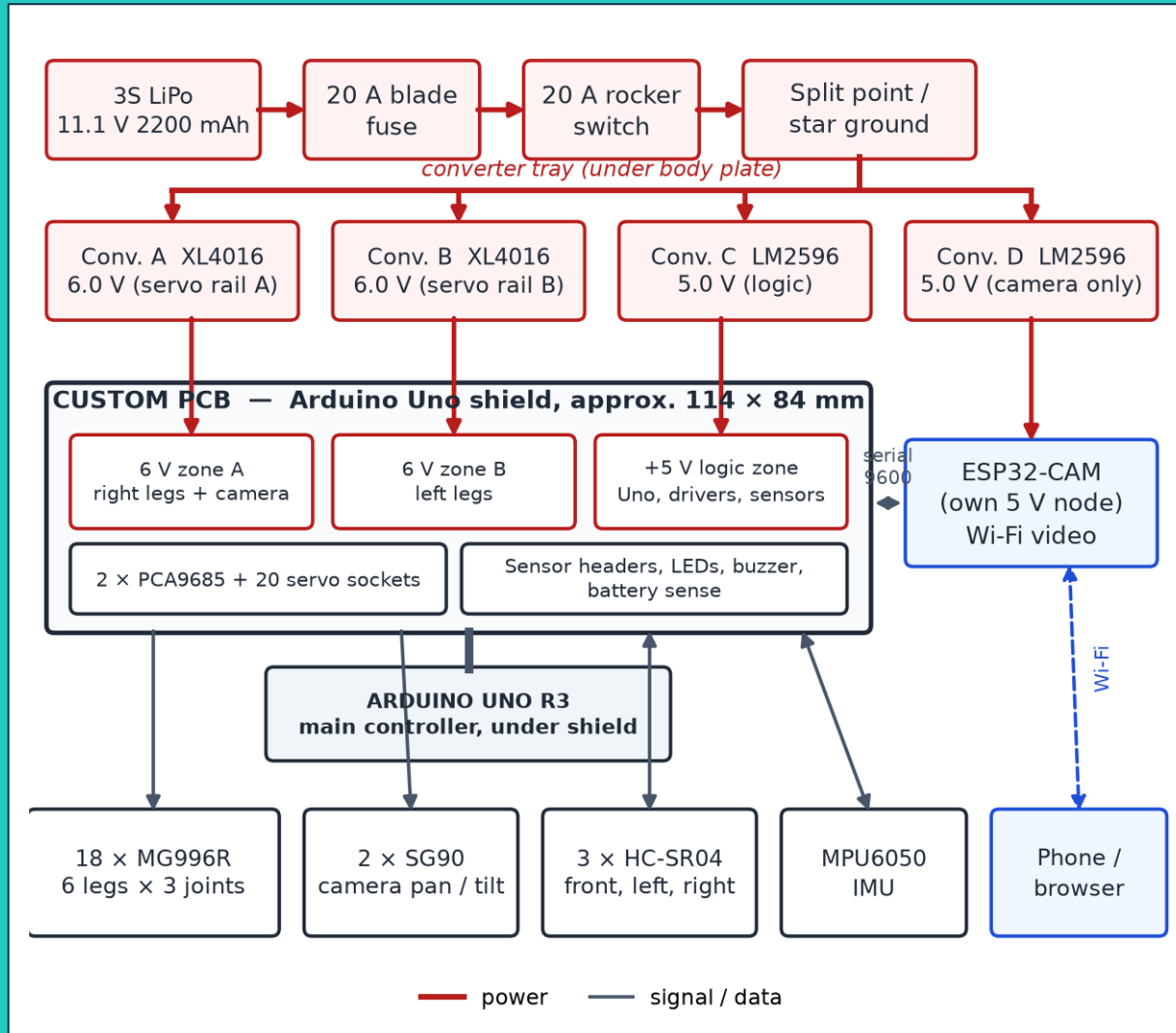
Support polygon and stability margin during a tripod gait



METHODOLOGY

- **Phase 1 — Architecture & PCB design:** multi-rail power distribution finalised; custom PCB shield designed in KiCad. *(Completed)*
- **Phase 2 — Mechanical fabrication:** 3D printing of the legs, body plate, electronics box and camera mount. *(Next)*
- **Phase 3 — Assembly & integration:** PCB soldering; servos, sensors and ESP32-CAM integrated step by step. *(Planned)*
- **Phase 4 — Kinematics & gait:** inverse kinematics for 18 joints on the Arduino Uno; tripod gait based on the static stability margin. *(Planned)*
- **Phase 5 — Command loop:** web page → ESP32-CAM serial link → Arduino gait engine → PCA9685 (I²C) → MG996R servos. *(Planned)*
- **Phase 6 — Safety & testing:** stop forward motion if an obstacle is within 30 cm; tilt and low-battery checks; calibration and testing. *(Planned)*

SYSTEM ARCHITECTURE



Planned system architecture of the hexapod

- **Arduino Uno R3** is the only controller: gait, sensor reading and joint commands.
- **ESP32-CAM** is a camera peripheral: video goes to the phone over Wi-Fi; single-character commands reach the Uno over a serial link.
- **Power:** 3S LiPo → 20 A fuse → switch → four buck converters.
- **Rails:** two 6 V servo rails (XL4016), a 5 V logic rail and a separate 5 V camera rail (LM2596).
- **One custom PCB shield** carries all power zones, driver modules and connectors, with a single star-ground point.



HARDWARE COMPONENTS & TOOLS USED

- **Controller & camera:** Arduino Uno R3; ESP32-CAM (OV2640).
- **Actuation:** $18 \times$ MG996R (legs), $2 \times$ SG90 (camera pan/tilt), $2 \times$ PCA9685 PWM drivers.
- **Sensing:** $3 \times$ HC-SR04 ultrasonic sensors, MPU6050 IMU.
- **Power:** 3S LiPo 2200 mAh (two packs), $2 \times$ XL4016 (6 V), $2 \times$ LM2596 (5 V), 20 A fuse and switch.
- **Software:** KiCad, Arduino IDE, VS Code, Git/GitHub.
- **Cost:** estimated purchase cost $\approx$ ₹17,346 (build-plan estimate); 3D printing provided by the college.
- **Status:** all required components have been procured; none installed yet.



Some of the procured components



PLANNED IMPLEMENTATION — CONTROL

- **Inverse kinematics:** foot position (x, y, z) converted into coxa–femur–tibia joint angles (θ_1 , θ_2 , θ_3) for each leg.
- **Tripod gait:** the six legs moved in two alternating groups of three (swing and stance).
- **PWM offloading:** two PCA9685 boards (I²C addresses 0x40 and 0x41) generate the 50 Hz servo pulses in hardware.
- **IMU posture correction:** MPU6050 pitch and roll used to correct the body posture on slopes.
- **Ultrasonic interlock:** front, left and right HC-SR04 sensors checked in the main loop to override forward motion.
- **Calibration & memory:** servos centred (PWM count $\approx$ 307) before fitting horns; gait tables stored in PROGMEM to save the Uno's 2 KB RAM.

These features are part of the planned design and will be verified during integration and testing.



PLANNED IMPLEMENTATION — POWER & HARDWARE

- **Multi-rail power:** XL4016 converters for the servo rails and LM2596 for logic, keeping servo current away from the logic circuits.
- **Camera supply:** a dedicated converter and $\geq 1000 \mu\text{F}$ capacitors to absorb the ESP32-CAM's $\sim 700 \text{ mA}$ Wi-Fi bursts.
- **Servo master-disable:** PCA9685 OE line pulled HIGH at boot, so servos stay off until a safe pose is written.
- **Battery monitoring:** resistor divider on A0; warning at 10.5 V and safe sit-down at 9.9 V.
- **Star ground:** all negatives meet at one point to avoid ground loops and I²C errors.
- **Mechanical:** converters in an open, slotted tray under the body plate for cooling; brass heat-set inserts in 3D-printed parts.

These features are part of the planned design and will be verified during integration and testing.



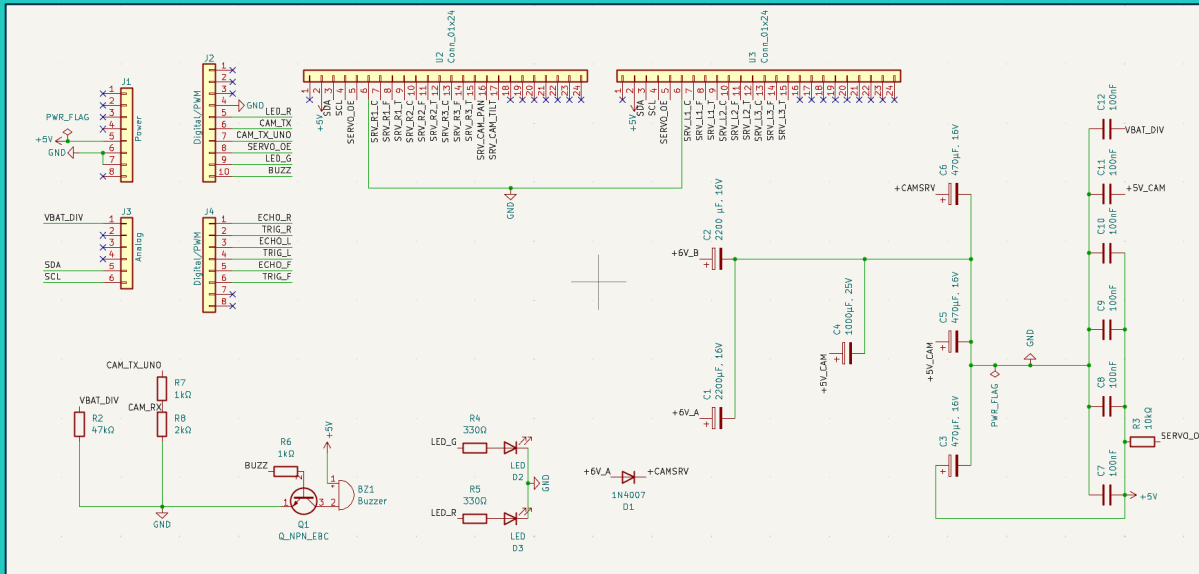
DESIGN COMPARISON: CONVENTIONAL vs PROPOSED

ASPECT	TYPICAL HOBBY BUILD	OUR DESIGN APPROACH
Wiring	Breadboard and ~90 loose jumper wires	Custom PCB shield with labelled sockets
Servo PWM	Pulses generated by the Arduino	Two PCA9685 boards generate pulses in hardware
Servo power	Single 5 V rail, often from the Arduino	Two 6 V XL4016 rails through PCB copper zones
Camera power	Shares the logic supply	Dedicated converter with bulk capacitors
Protection	Often none	20 A fuse, rocker switch, OE interlock, battery sensing
Cooling	Regulators inside a closed box	Open, slotted converter tray under the body plate
Mobility	Wheels stop at rubble and gaps	18-DOF legs; thigh load ≈ 47 % of servo stall torque (design estimate)

Expected benefits are based on design calculations and will be confirmed through testing after assembly.

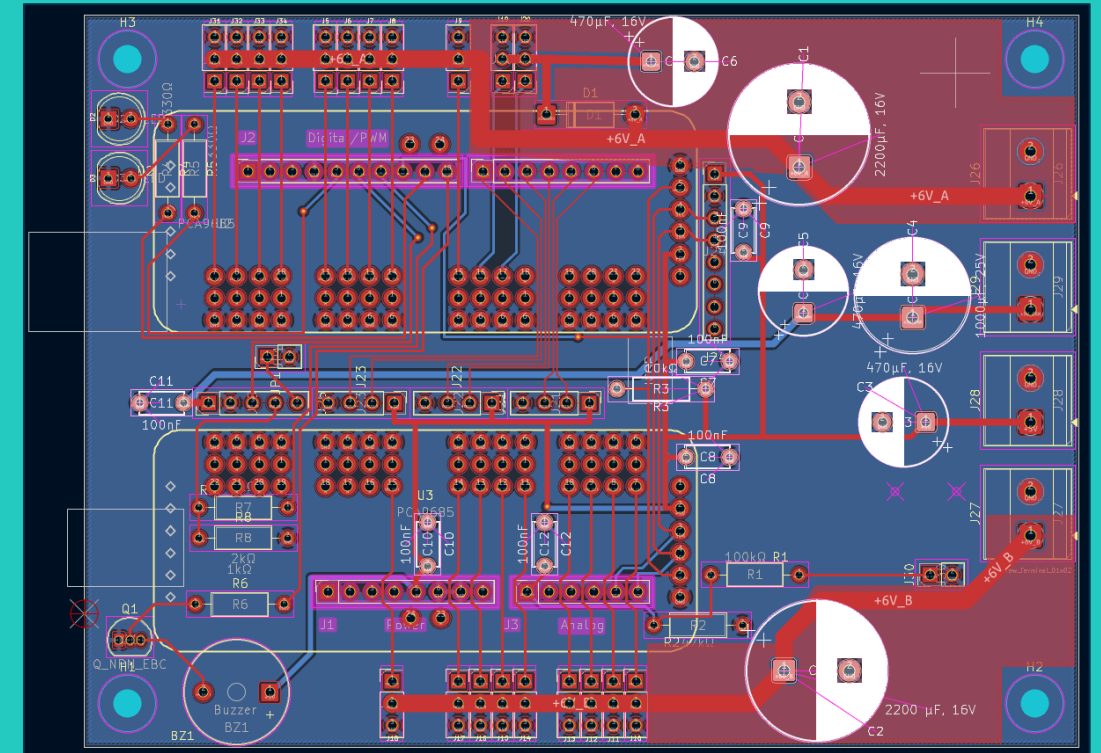
CUSTOM PCB DEVELOPMENT (1/2)

Schematic ✓ → PCB Layout ✓ → 3D View ✓ → Gerber Generation ✓ → Fabrication: Pending



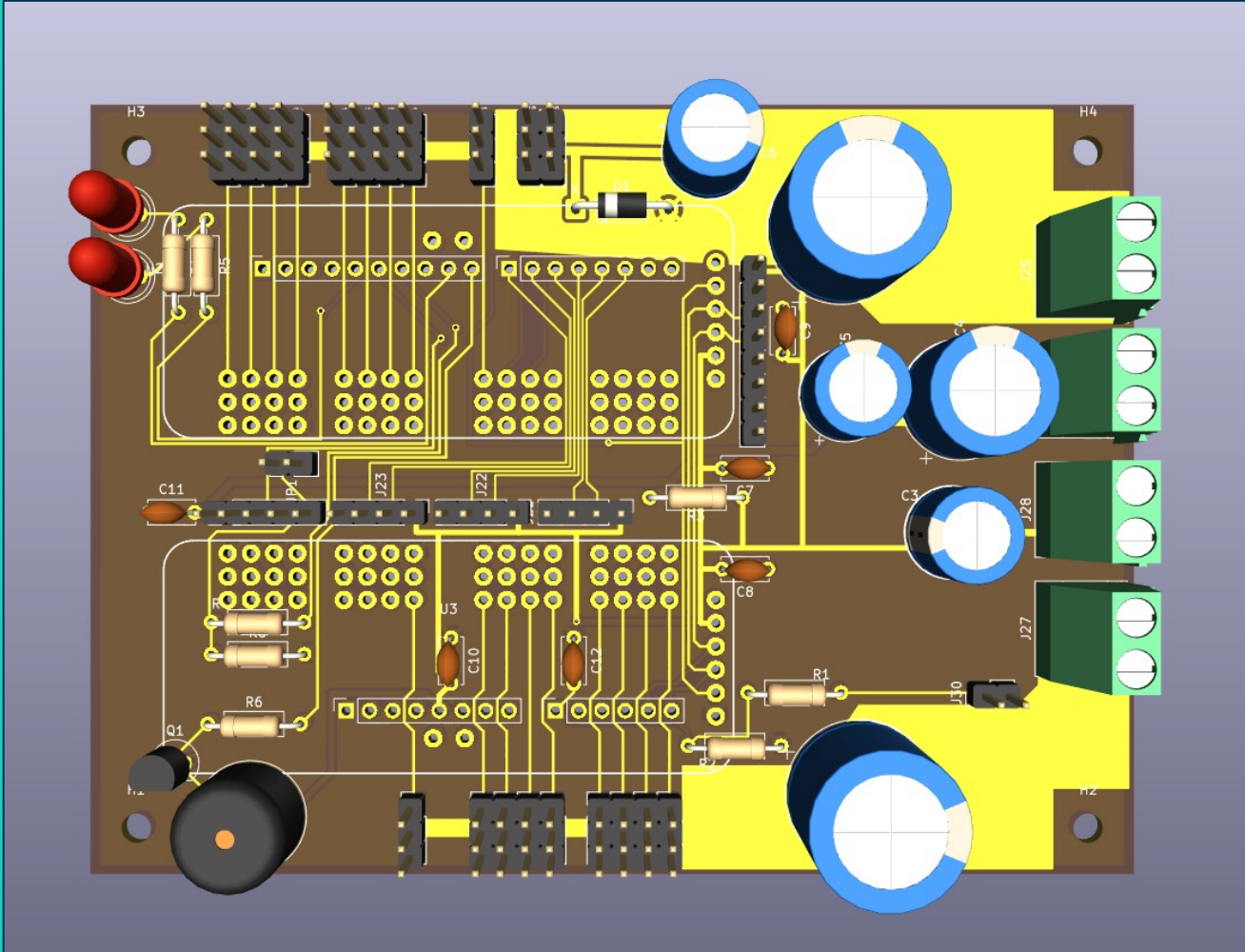
Schematic in KiCad (sheet 1 of 2)

- Shield for the Arduino Uno, about 114×84 mm, two copper layers.
- Separate zones: 6 V zone A (right legs + camera servos), 6 V zone B (left legs), 5 V logic and an isolated 5 V camera node.
- 20 labelled servo sockets, sensor headers and PCA9685 module sockets.



PCB layout in KiCad

CUSTOM PCB DEVELOPMENT (2/2)



3D view of the PCB in KiCad

Design checks

- ERC: 0 errors, 0 warnings.
- DRC: 0 unconnected items; remaining markers are silkscreen clearance (printed labels only).

Gerber files generated (27 Sep 2026)

- Front and back copper, solder mask, silkscreen and paste layers.
- Board outline (Edge Cuts), PTH and NPTH drill files, Gerber job file.

Status

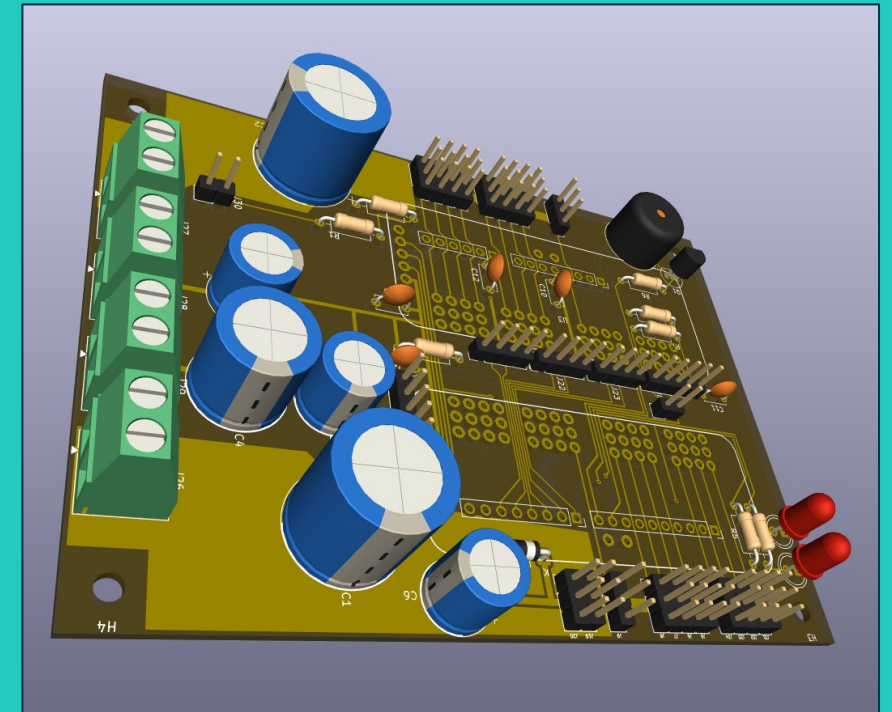
PCB design complete and Gerber files ready.

Fabrication / ordering is the next pending step — the board has not been manufactured yet.



CURRENT PROJECT STATUS

STAGE	STATUS
Planning & research	✓ Completed
Component selection & procurement	✓ Completed
PCB schematic & layout	✓ Completed
Design checks (ERC / DRC)	✓ Completed
Gerber file generation	✓ Completed
PCB ordering / fabrication	Pending — next step
3D printing of mechanical parts	Next
Mechanical assembly & PCB installation	Planned
Electronics, sensor & camera integration	Planned
Programming, calibration & testing	Planned



PCB design — ready for fabrication



FUTURE PLAN OF ACTION

Month	Activity	Status
AUG	Literature review, base-paper study and architecture finalisation	Completed
SEP	Component procurement; PCB schematic, layout, design checks and Gerber generation	Completed
OCT	Order and fabricate the PCB; start 3D printing of legs and body parts	To be done
NOV	Assemble the mechanical structure; mount servos and electronics; install the PCB	To be done
DEC	Integrate sensors and ESP32-CAM; develop gait control; calibrate the system	To be done
JAN	Test walking, stability, obstacle sensing and surveillance; final demonstration	To be done

Plan is tentative and depends on PCB delivery time and 3D-printer availability.



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THANK YOU!